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(b) System device power = 30 W, 24/7 applicability.

(c) Night time operational power at PR3 = (20 × 22) + (30) = **470 W**.

(d) Day time operational power at PR3 = (20 × 2) + (30) = **70 W**.

(e) Night time operational power at PR5 = (20 × 12) + (30) = **270 W**.

(f) Day time operational power at PR5 = (20 × 2) + (30) = **70 W**.

(g) Night time duration of operation over a year = 365 days × 24 h × 0.5 = 4380 h/year.

(h) Day time duration of operation over a year = 365 days × 24 h × 0.5 = 4380 h/year.

The annual energy consumption indicator (AECI) calculation is—

$$D_E = \left[(470\times4380\times0.5)+(270\times4380\times0.5)+(70\times4380)\right]/18\ 000 = 107.07\,\text{Wh.m}^{-2} \qquad \text{F12}$$

where

D_E = annual energy consumption indicator (Wh.m^{-2})

P_j = operational power. Three states = PR3 470 W, PR5 270 W, day time 70 W

t_j = duration of operation over a year. Day time 4380 h, night time 4380 h

A = size of the area lit = 18 000 m^2

m = number of periods with different operational power = 3

Therefore, $D_E = \left[(470\times4380)+(270\times4380)+(70\times4380)\right]/18\,000 = 197.10\ Wh.\,m^{-2}$.

F.3.2.2 Worked example with constant light output (CLO)

Example lighting scheme from Paragraph F2.5 above, with added lighting scheme operational parameters (numerical values are indicative only):

(a) Light point controller power = 2 W, 24/7 applicability.

(b) System device power = 30 W, 24/7 applicability.

(c) Night time operational power at PR3 = (20 × 19) + (30) = **410 W**.

(d) Day time operational power at PR3 = (20 × 2) + (30) = **70 W**.

(e) Night time operational power at PR5 = (20 × 10.5) + (30) = **240 W**.

(f) Day time operational power at PR5 = (20 × 2) + (30) = **70 W**.

(g) Night time duration of operation over a year = 365 days × 24 h × 0.5 = 4380 h/year.

(h) Day time duration of operation over a year = 365 days × 24 h × 0.5 = 4380 h/year.

The annual energy consumption indicator (AECI) calculation is—

$$D_E = \left[(410\times4380\times0.5)+(240\times4380\times0.5)+(70\times4380)\right]/18\ 000 = 96.12\,\text{Wh.m}^{-2} \qquad \text{F13}$$

where

D_E = annual energy consumption indicator (Wh.m^{-2})

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